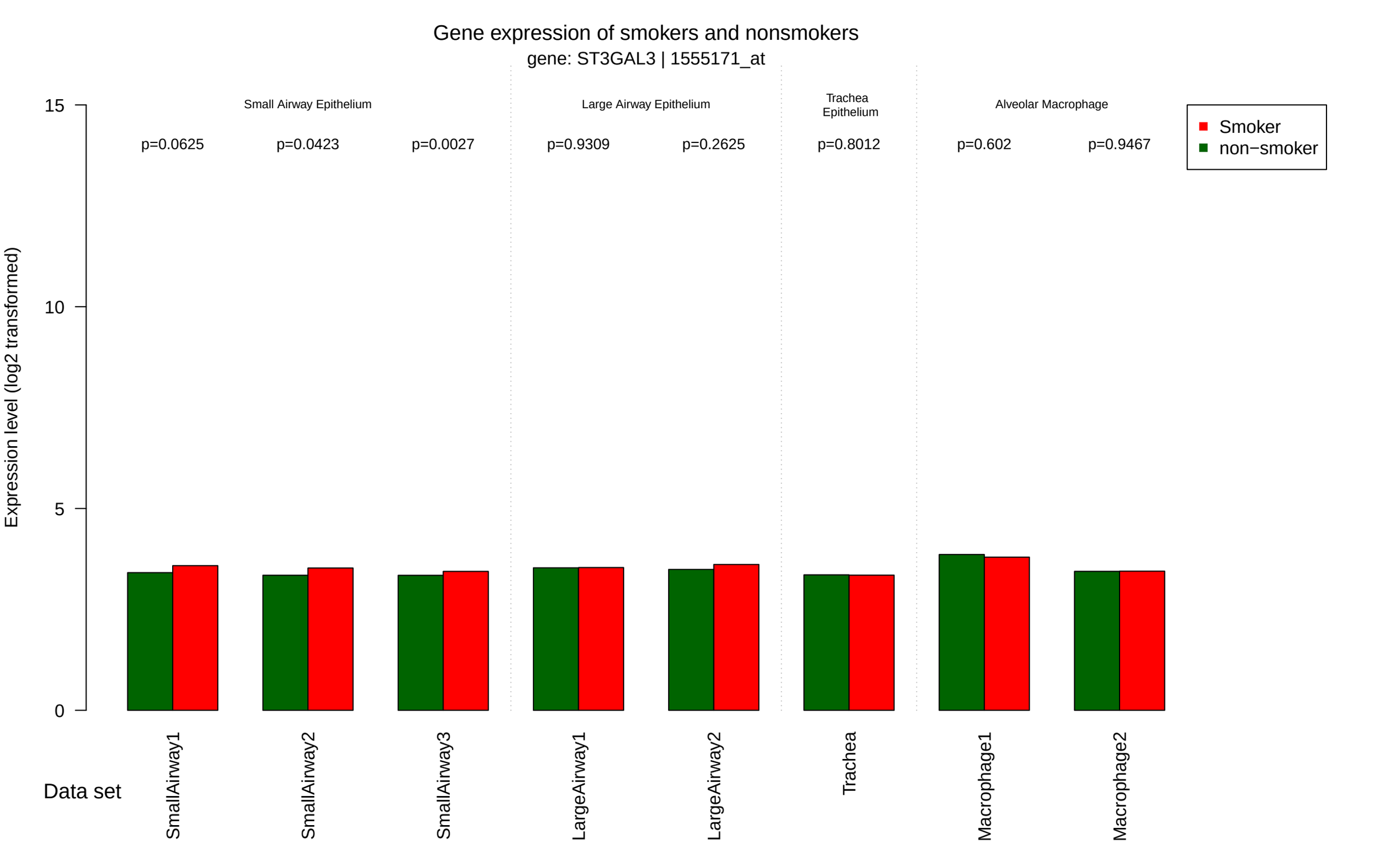
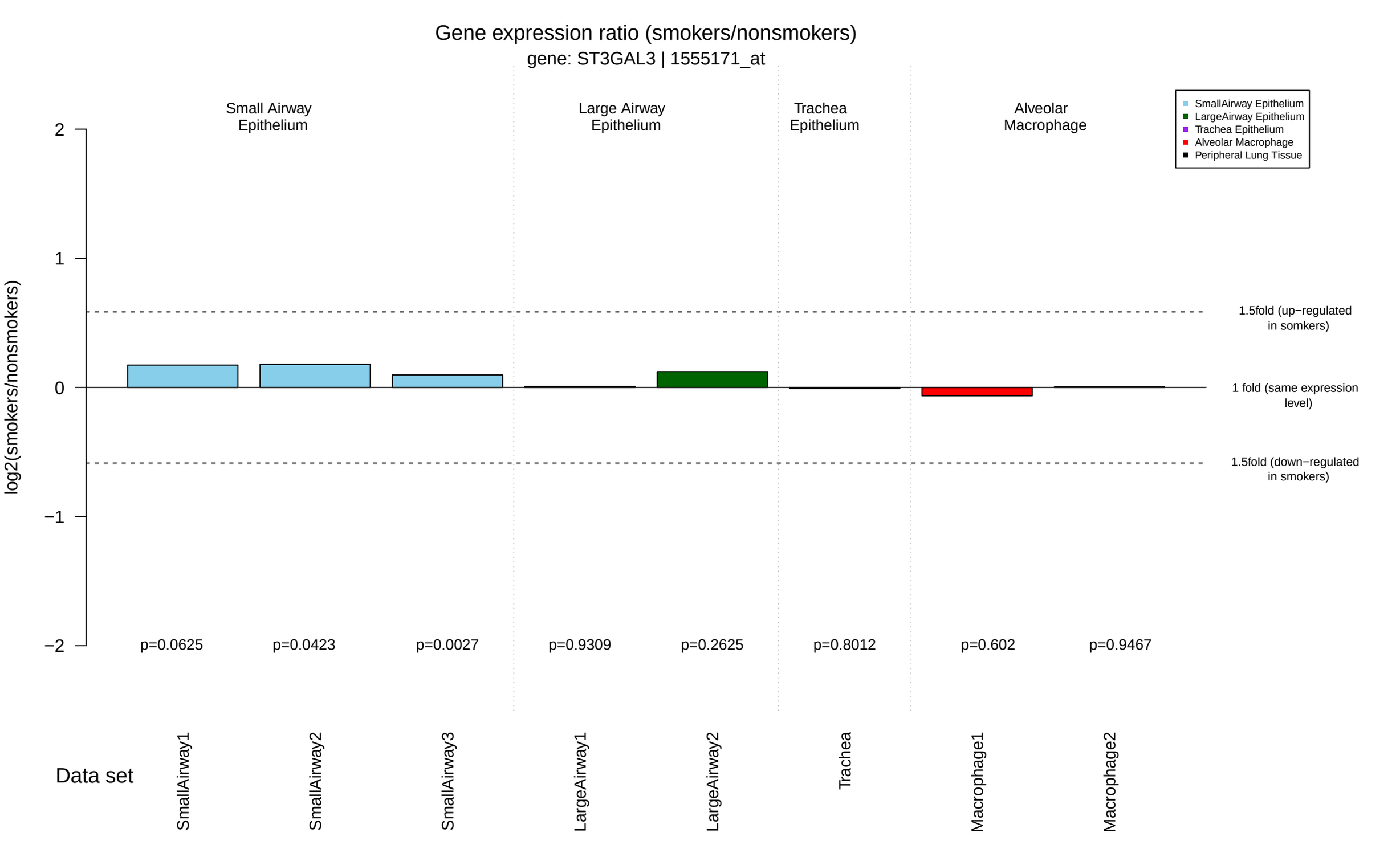
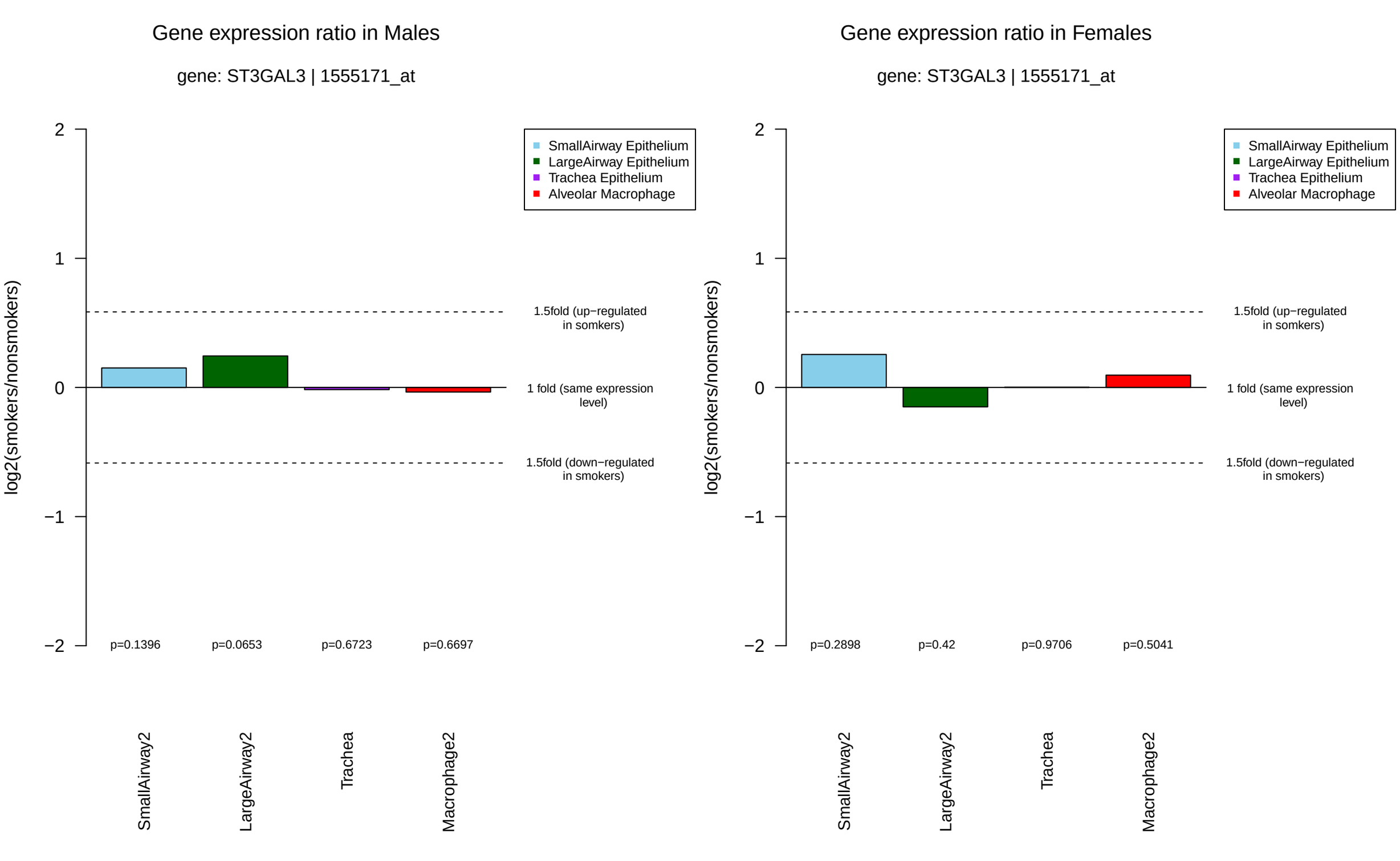
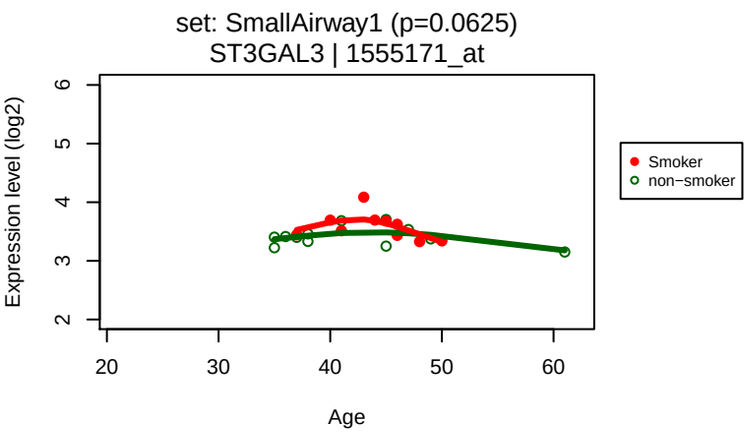
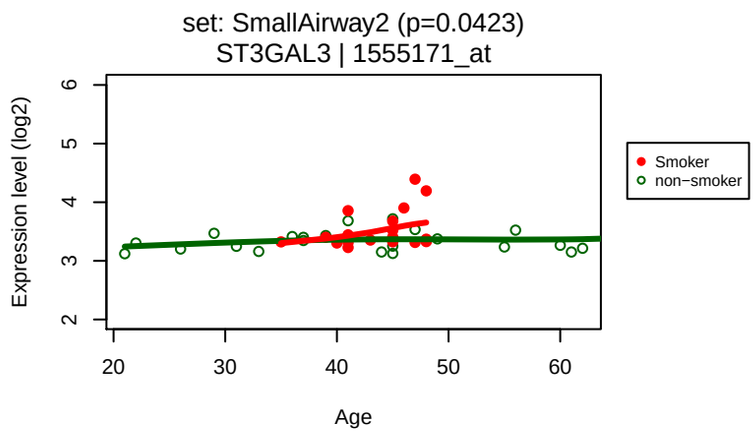
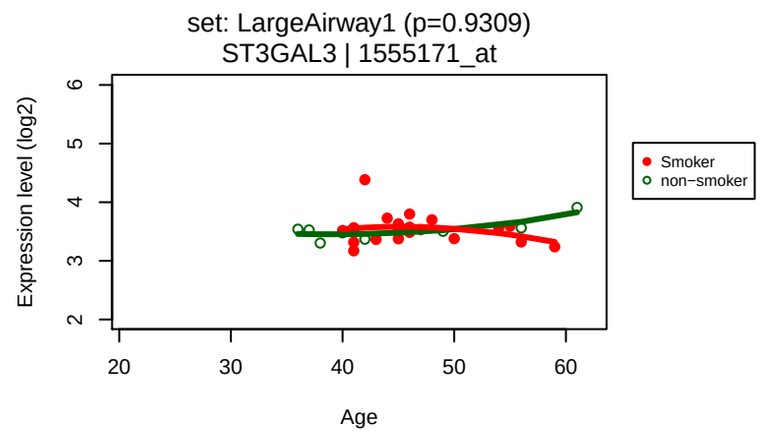

Gene expression of smokers and nonsmokers
gene: ST3GAL3 | 1555171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0625
p=0.0423
p=0.0027
p=0.9309
p=0.2625
p=0.8012
p=0.602
p=0.9467
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

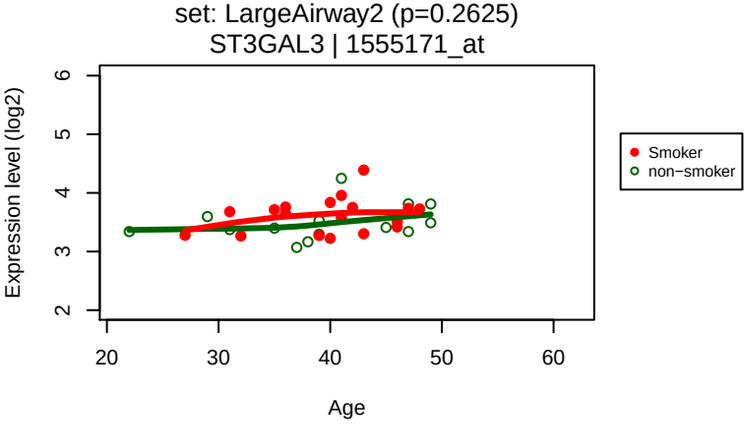

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL3 | 1555171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0625
p=0.0423
p=0.0027
p=0.9309
p=0.2625
p=0.8012
p=0.602
p=0.9467
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

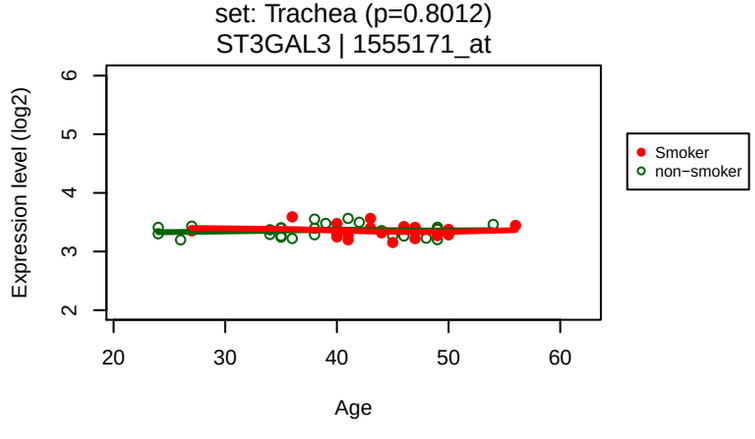

Gene expression ratio in Males
gene: ST3GAL3 | 1555171_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1396
p=0.0653
p=0.6723
p=0.6697
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL3 | 1555171_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2898
p=0.42
p=0.9706
p=0.5041
SmallAirway2
LargeAirway2
Trachea
Macrophage2

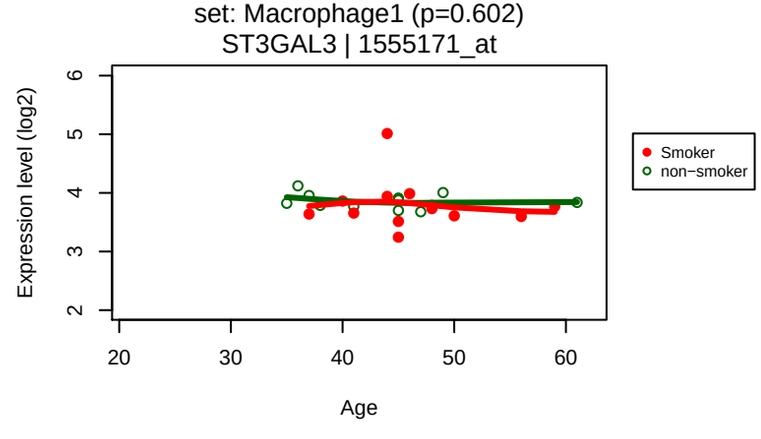

set: SmallAirway1 (p=0.0625)
ST3GAL3 | 1555171_at
Expression level (log2)
Age
Smoker
non−smoker

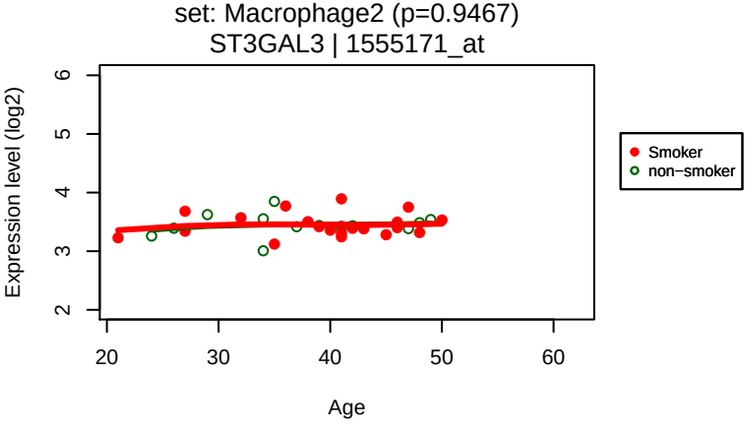

set: SmallAirway2 (p=0.0423)
ST3GAL3 | 1555171_at
Expression level (log2)
Age
Smoker
non−smoker

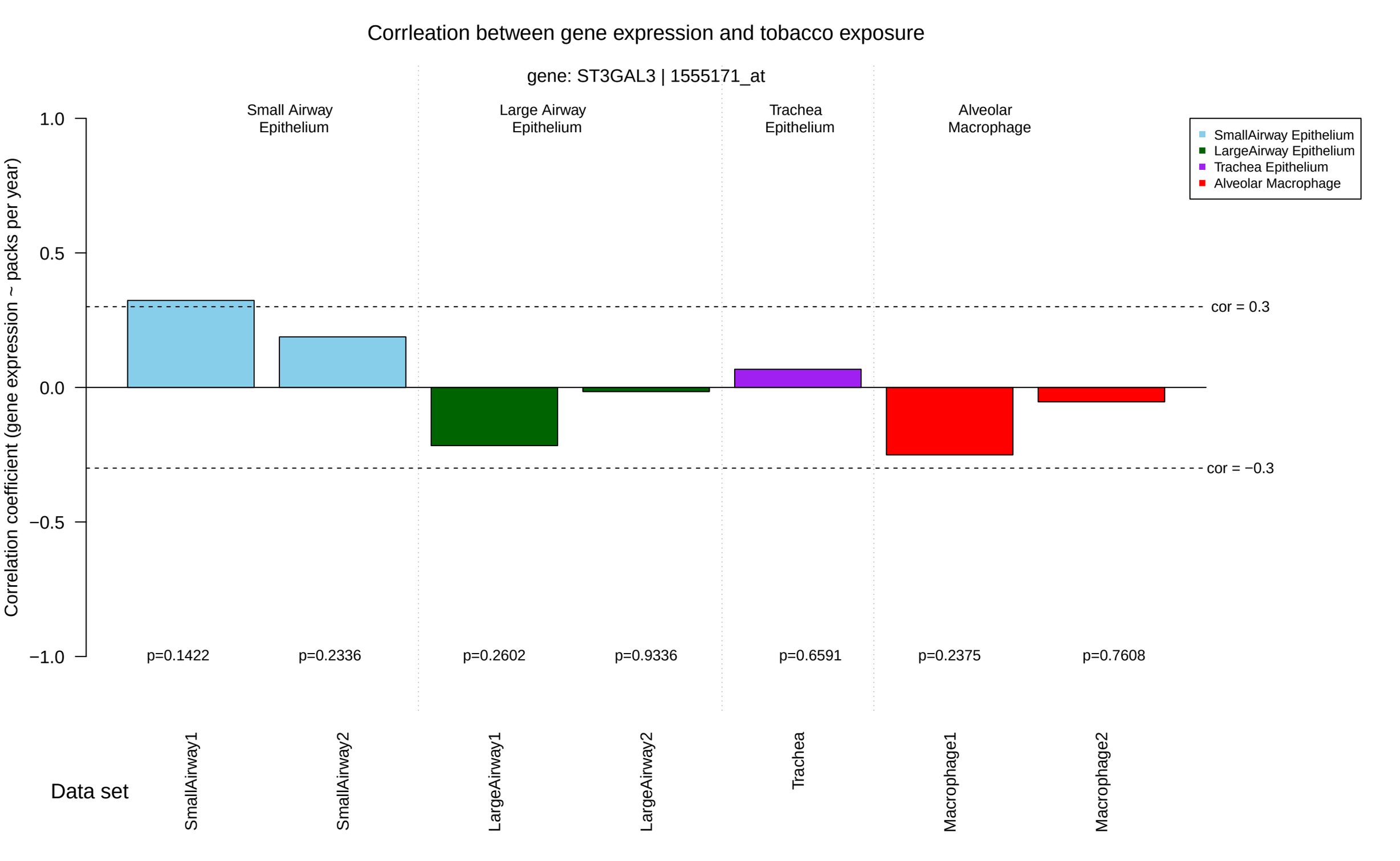

set: LargeAirway1 (p=0.9309)
ST3GAL3 | 1555171_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2625)
ST3GAL3 | 1555171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8012)
ST3GAL3 | 1555171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.602)
ST3GAL3 | 1555171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9467)
ST3GAL3 | 1555171_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL3 | 1555171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1422
p=0.2336
p=0.2602
p=0.9336
p=0.6591
p=0.2375
p=0.7608
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2